Offer #2026-09732

Doctorant F/H De quoi rêvent les GNN ? Méthode d'interprétabilité basée sur l'extraction de motifs

Contract type : Fixed-term contract

Level of qualifications required : Graduate degree or equivalent

Fonction : PhD Position

About the research centre or Inria department

The Inria Rennes - Bretagne Atlantique Centre is one of Inria's eight centres and has more than thirty research teams. The Inria Center is a major and recognized player in the field of digital sciences. It is at the heart of a rich R&D and innovation ecosystem: highly innovative PMEs, large industrial groups, competitiveness clusters, research and higher education players, laboratories of excellence, technological research institute, etc.

Context

Within the framework of a partnership public with French National Research Agency (ANR)

Assignment

Context

Graph Neural Networks (GNNs) [1] have become increasingly popular in recent years due to their ability to process graph-structured data (e.g., social networks, molecules, knowledge graphs, etc.). These models have set the new state-of-the-art in tasks such as link prediction and graph classification, achieving impressive results compared to previous approaches. However, like other neural network-based approaches, GNNs suffer from a lack of interpretability: it is nearly impossible for a human expert to understand the reasoning behind a GNN's decision.

While numerous explainability (XAI) methods have been proposed for traditional neural networks (e.g., those processing text or images) [2,3], very few works have focused on the case of graph data. In [4], a GNN explainability method based on pattern mining [5] was proposed. This method is unique in that it directly leverages the components activated during the GNN's decision-making process, extracting patterns called ``activation rules''. These activation rules are then linked to the input graph data, enabling the generation of explanations in the form of subgraphs.

This preliminary method, however, has several limitations. First, the activation rules are only extracted for a single layer of the GNN, limiting their expressiveness. Additionally, non-activations are not taken into account, even though they can be crucial for explaining a decision. Another key limitation is the overwhelming number of activation rules generated, due to the combinatorial nature of the extraction process. A quality measure is therefore needed to select a relevant subset of rules. The current method uses a statistical-based measure on the rule set, but it fails to link the rules to the corresponding parts of the input graphs.

Thesis Objectives

The goal of this PhD thesis is to provide human users with rich, precise, and understandable explanations of GNN decisions. The first work will be on improving the expressiveness of the activation rules extracted from the GNN. The aim is to develop a method that extracts activation rules from components across multiple GNN layers, taking into account both activations and non-activations (e.g., negative patterns [6]).

As this leads to an extremely large search space of potential rules, the proposed approach must return a small, relevant subset of rules that best explain the GNN's decision. To achieve this, techniques based on Information Theory—particularly the Minimum Description Length (MDL) principle [7]—will be explored.

A final theoretical contribution will be the development of methods to ``translate'' these expressive activation rules into the input graph space, in order to provide comprehensible explanations grounded in the original data. Targeted applications include molecular graphs and semantic web knowledge graphs. A promising direction for achieving a robust mapping between activation rules and input graphs is to leverage the knowledge stored in Large Language Models (LLMs) to capture domain-specific semantics of the graph data.

[1] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, G. Monfardini. The Graph Neural Network Model. In IEEE Transactions on Neural Networks, vol. 20, no. 1, pp. 61-80 (2009).

[2] M. Tu?lio Ribeiro, S. Singh, C. Guestrin. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. KDD 2016: 1135-1144

[3] Scott M. Lundberg, Su-In Lee: A Unified Approach to Interpreting Model Predictions. NIPS 2017: 4765-4774

[4] L. Veyrin-Forrer, A. Kamal, S. Duffner, M. Plantevit, C. Robardet. On GNN explainability with activation rules. Data Min Knowl Disc (2022).

[5] C. Aggarwal, J. Han. Frequent Pattern Mining. Springer, Cham (2014).

[6] T. Guyet, R. Quiniou. NegPSpan: efficient extraction of negative sequential patterns with embedding constraints. Data Min. Knowl. Discov. 34(2): 563-609 (2020)

[7] P. Grünwald. The Minimum Description Length Principle. The MIT Press (2007)

Main activities

Main activities

- Develop programs
- Design experimental platforms
- Write papers
- Test, change up until validation
- Distribute the work via publications and talks
- Present the works' progress to partners

Skills

The candidate should have a strong interest in machine learning in general, with a particular focus on neural networks, statistics, algorithms, and programming.

Benefits package

- Subsidized meals
- Partial reimbursement of public transport costs
- Leave: 7 weeks of annual leave + 10 extra days off due to RTT (statutory reduction in working hours) + possibility of exceptional leave (sick children, moving home, etc.)
- Possibility of teleworking (after 6 months of employment) and flexible organization of working hours
- Professional equipment available (videoconferencing, loan of computer equipment, etc.)
- Social, cultural and sports events and activities
- Access to vocational training

Remuneration

monthly gross salary 2300 euros

General Information

- **Theme/Domain :** Data and Knowledge Representation and Processing Information system (BAP E)
- **Town/city :** Rennes
- **Inria Center :** Centre Inria de l'Université de Rennes
- **Starting date :** 2026-03-02
- **Duration of contract :** 3 years, 10 months
- **Deadline to apply :** 2026-02-28

Contacts

- **Inria Team :** LACODAM
- **PhD Supervisor :**
  Cellier Peggy / Peggy.Cellier@irisa.fr

About Inria

Inria is the French national research institute dedicated to digital science and technology. It employs 2,600 people. Its 200 agile project teams, generally run jointly with academic partners, include more than 3,500 scientists and engineers working to meet the challenges of digital technology, often at the interface with other disciplines. The Institute also employs numerous talents in over forty different professions. 900 research support staff contribute to the preparation and development of scientific and entrepreneurial projects that have a worldwide impact.

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Instruction to apply

Please submit online : your resume, cover letter and letters of recommendation eventually

Defence Security :
This position is likely to be situated in a restricted area (ZRR), as defined in Decree No. 2011-1425 relating to the protection of national scientific and technical potential (PPST).Authorisation to enter an area is granted by the director of the unit, following a favourable Ministerial decision, as defined in the decree of 3 July 2012 relating to the PPST. An unfavourable Ministerial decision in respect of a position situated in a ZRR would result in the cancellation of the appointment.

Recruitment Policy :
As part of its diversity policy, all Inria positions are accessible to people with disabilities.